WASP-11b

R-0781

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-10_221224` · created 2026-07-28T06:14:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459937.6511 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.048
Transit depth (Rp/Rs)^2	2.07 +/- 1.39 [%]
Orbital Inclination (inc)	85.11 +/- 1.74
Airmass coefficient 1 (a1)	841.14 +/- 2950.98
Airmass coefficient 2 (a2)	-3.0 +/- 0.012
Scatter in the residuals of the lightcurve fit is	8868.73 %
Best Comparison Star	#1 - [538, 149]
Optimal Aperture	3.0
Optimal Annulus	3.0
Transit Duration (day)	0.035 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.048 vs 0.131 (deviation 0.0σ)
Duration fit vs published	0.035 d vs 0.1075 d (deviation 0.03σ)
Mid-time O–C	-6.9 min
Depth SNR	0.0
Residual scatter	8868.73 %

Light curve

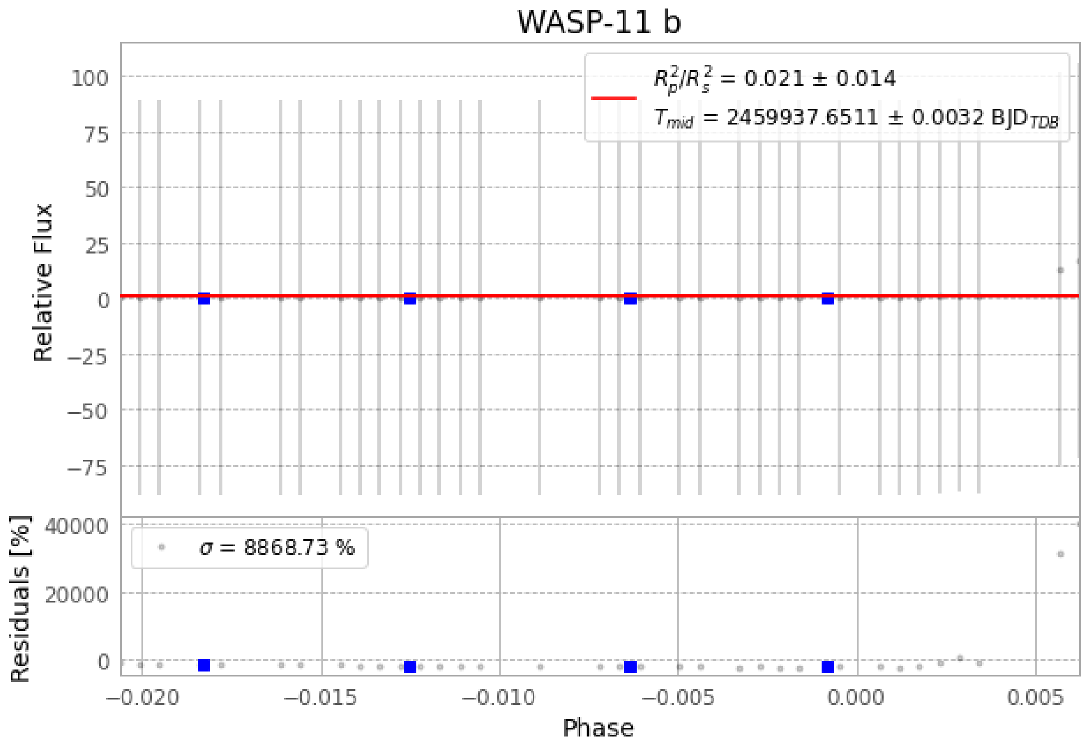

Figure 1 — FinalLightCurve_WASP-11 b_2022-12-23.png (SHA-256 b360d886b797f4ac…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [519, 410], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 fa6cd1101c9e13bf…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

89 FITS frames (58 MB), `HATP-10221224013015.FITS` → `HATP-10221224055416.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-221224.log (666e77afb409…), FinalLightCurve_WASP-11 b_2022-12-23.png (b360d886b797…), FinalLightCurve_WASP-11 b_2022-12-23.csv (0aa7b60606c7…)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)